Math Inquirers

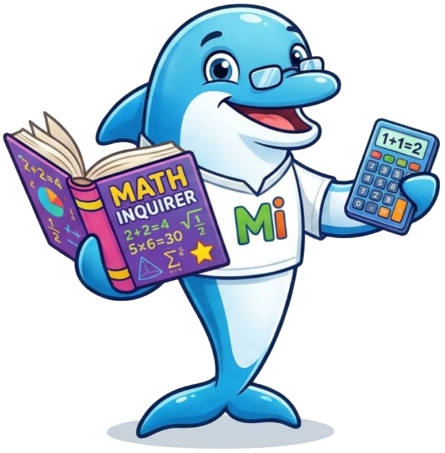
Think. Question. Prove.

Level 1 (ages 6-8) - Term 2

Tutor Answer Key

Book 4

Lessons 16 to 20



FOR THE TUTOR AND GROWN-UPS ONLY - keep away from the child

Contents

Math Inquirers Level 1, Term 2 - Tutor Answer Key, Book 4

Lesson 16 **Skip-Count Tables** page 3
Multiplication: Skip-Count Tables

Lesson 17 page 5

Lesson 18 **Times Stories** page 8
Multiplication And Division: Times Stories

Lesson 19 **Picture Equations** page 11
Reasoning About Equality: Picture Equations

Lesson 20 **Find One First** page 14
Division And Money: Find One First

Answer key: Lesson 6, Skip-Count Tables

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Count in 5s. Fill in the missing numbers.

Answer: 15, 25, 35

Counting in 5s: 5, 10, 15, 20, 25, 30, 35, 40. The missing numbers are 15, 25 and 35.

Watch for: Writing 14 or 16 (counting in 1s).; Skipping a number.

S2. A frog jumps along the number line. Fill in the blanks.

Answer: 3 jumps; each jump is 5; 3 × 5 = 15 (5 × 3 = 15 is also right)

The frog makes 3 jumps and each jump is 5. 5 + 5 + 5 = 15, so 3 jumps of 5 is 3 × 5 = 15.

Watch for: Counting the numbers on the line instead of the jumps.; Writing 3 + 5 = 8.

S3. Ravi has 4 bundles of 10 straws. How many straws does Ravi have in all?

Answer: 10 + 10 + 10 + 10 = 40; 4 × 10 = 40

4 bundles of 10: 10 + 10 + 10 + 10 = 40. So 4 × 10 = 40 and Ravi has 40 straws.

Watch for: Writing 4 + 10 = 14.; Counting the bundles as 4 straws.

S4. Each bike has 2 wheels. How many wheels do 6 bikes have?

Answer: 12 wheels (6 × 2 = 12; 2 × 6 = 12 is also right)

6 bikes with 2 wheels each: 2 + 2 + 2 + 2 + 2 + 2 = 12. So 6 × 2 = 12.

Watch for: Writing 8 (6 + 2).; Losing count of the bikes.

S5. Fill in the table.

Answer: 10, 20, 30, 40

2 × 5 = 10, 4 × 5 = 20, 6 × 5 = 30 and 8 × 5 = 40.

Watch for: Adding: 2 + 5 = 7.; Writing the number of jumps in the Total box.

S6. Read each statement. Decide if it is True or False.

Answer: True, False, True, False

5 × 4 = 20 is true. 3 × 10 is 30, not 40: false. 6 × 2 = 12 is true. 8 × 5 is 40, not 35: false.

Watch for: Marking 3 × 10 = 40 as true.; Marking 8 × 5 = 35 as true because it ends in 5.

Think harder (page 3)

T1. Look at the dots. The top 2 rows are shaded. Fill in the blanks.

Answer: 2 × 5 = 10; 4 × 5 = 20

2 rows of 5 dots is 2 × 5 = 10. All 4 rows are two lots of 10 dots: 10 + 10 = 20, so 4 × 5 = 20.

Watch for: Writing 10 for both.; Adding 4 + 5 = 9.

T2. Colour the 2s. Ring the 5s. Then fill in the blanks.

Answer: 10, 20, 30, 40

The 2s are 2, 4, 6 ... 40. The 5s are 5, 10, 15 ... 40. The numbers in both are 10, 20, 30 and 40.

Watch for: Listing every number in the 5s.; Missing 40.

T3. Is each number in the 5s table? Circle Yes or No.

Answer: 50 Yes; 27 No; 65 Yes

50 ends in 0: yes. 27 ends in 7: no. 65 ends in 5: yes.

Watch for: Saying 27 is in the table because it is close to 25.; Saying 65 is not in the table because it is a big number.

Go further (page 4)

G1. Here are the 4s up to 40. Ring each number that is also in the 5s table. Then fill in the blanks.

Answer: 20 and 40

The 4s are 4, 8, 12, 16, 20, 24, 28, 32, 36, 40. The ones that end in 5 or 0 are 20 and 40.

Watch for: Ringing 16 or 24 (they do not end in 0 or 5).; Ringing only 20.

G2. Count in 3s up to 30. Fill in the missing numbers. Then find the numbers in both the 3s and the 5s.

Answer: 12, 18, 24; 15 and 30

The 3s are 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. The 5s up to 30 are 5, 10, 15, 20, 25, 30. The numbers in both are 15 and 30.

Watch for: Writing 11 or 13 (counting wrongly after 9).; Writing 10 for a number in both tables.

G3. Ali jumps 5 each time. Mei jumps 4 each time. They both want to reach 40.

Answer: Ali 8 jumps, Mei 10 jumps, 2 more

8 jumps of 5 make 40 (8 × 5 = 40). 10 jumps of 4 make 40 (10 × 4 = 40). 10 - 8 = 2 more jumps.

Watch for: Guessing the number of jumps of 4 without counting in 4s.; Forgetting to subtract at the end.

Investigation (page 5): Last digits

10 and 20 are in both tables (15 and 25 are only in the 5s). Both end in 0. The 5s end in 5 or 0 and the 2s end in 0, 2, 4, 6 or 8: the only digit in both lists is 0.

Look for: A child who tests several numbers, writes the last digits of both tables and sees that only 0 is in both lists. Credit any clear reason. Do not mark a child down for stopping at a few tests.

Exit ticket (page 5)

E1. Count in 5s. Fill in the missing numbers.

Answer: 15 and 25

Counting in 5s: 5, 10, 15, 20, 25, 30.

Watch for: Writing 11 or 12.; Skipping a number.

E2. Which sentence is true? Write its number in the bracket.

Answer: (2) 4 × 10 = 40

4 × 10 = 40 (4 jumps of 10). 3 × 5 is 15, not 20; 6 × 2 is 12, not 14; 8 × 5 is 40, not 35.

Watch for: Choosing 8 × 5 = 35.; Choosing 3 × 5 = 20.

E3. Ben puts 5 stickers on each of 6 pages. How many stickers does Ben have in all?

Answer: 6 × 5 = 30 (5 × 6 = 30 is also right); 30 stickers

6 pages with 5 stickers on each page: 6 jumps of 5 land on 30.

Watch for: Writing 6 + 5 = 11.; Writing 6 × 5 = 11.

Sneak-peek puzzles (not in the workbook)

1. 20
2. 10 and 20. Ali says every second number and Mei says every fifth number, and they both say 10 and 20.

Answer key: Lesson 7, Sharing and Grouping

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Share 8 dots equally among the 4 plates. Draw the dots. Then fill in the blank.

Answer: Each plate has 2 dots

Deal one dot at a time to each of the 4 plates. After 2 rounds all 8 dots are drawn. Each plate has 2 dots, and 4 groups of 2 is 8.

Watch for: Putting all the dots on one plate.; Writing 4, the number of plates.

S2. Mei packs 12 cookies into bags. She puts 3 cookies in each bag. Draw a ring around each group of 3. Then fill in the blank.

Answer: Mei needs 4 bags

12 cookies in groups of 3: ring 3, ring 3, ring 3, ring 3. That is 4 groups, so Mei needs 4 bags. Count in jumps of 3 to check: 3, 6, 9, 12.

Watch for: Writing 3, the number in each bag.; Ringing groups of different sizes.

S3. Ali has 14 sweets. He shares them equally among 2 friends. How many sweets does each friend get?

Answer: Each friend gets 7 sweets

Dealing 14 sweets to 2 friends one at a time gives 7 each. Check: 2 groups of 7 is 14.

Watch for: Writing 2, the number of friends.; Writing 14 or 16 by adding.

S4. Share the sweets equally among the friends. Fill in the table.

Answer: 4, 3, 4

8 sweets among 2 friends is 4 each. 12 sweets among 4 friends is 3 each. 20 sweets among 5 friends is 4 each. Check: 2 groups of 4 is 8, 4 groups of 3 is 12, 5 groups of 4 is 20.

Watch for: Writing the number of friends in the last column.; Giving 5 for the last row because it has a 5.

S5. Ben puts 20 pencils into pots. He puts 5 pencils in each pot. How many pots does Ben use?

Answer: Ben uses 4 pots

5, 10, 15, 20 is 4 jumps, so Ben uses 4 pots. Check: 4 groups of 5 is 20.

Watch for: Writing 5, the number in each pot.; Stopping at 15.

S6. Mei shares 12 stickers equally among 4 friends. Each friend gets 3 stickers.

Answer: 4 × 3 = 12 (3 × 4 = 12 is also right); Yes

4 friends with 3 stickers each is 4 groups of 3: 4 × 3 = 12. (3 × 4 = 12 is also right.) That is the 12 stickers Mei began with, so the answer matches.

Watch for: Writing 4 × 3 = 7 by adding.; Circling No because the numbers look different.

S7. Read each story. Circle Share or Group.

Answer: Share; Group

Lily deals the beads out equally to 3 friends, so it is Share (4 beads each). Jamie puts 6 beads in each box, so it is Group (2 boxes).

Watch for: Choosing Share because both stories have 12 beads.; Choosing by the number 3 or 6.

Think harder (page 3)

T1. Ali has 19 counters. He puts them into groups of 4.

Answer: 4 groups; 3 left over

4, 8, 12, 16 is 4 groups, and 16 counters are used. 19 - 16 = 3 counters are left. Three is not enough for another group of 4.

Watch for: Writing 5 groups by rounding up.; Forgetting the counters left over.

T2. Ben shares 20 biscuits equally with his 3 cousins. Aisha says 3 people share the biscuits.

Answer: No; 5 biscuits each

Ben shares with his 3 cousins, so Ben is one of the sharers: Ben and 3 cousins is 4 people, not 3. 20 biscuits among 4 people is 5 each, because 4 groups of 5 is 20. (A story that says 'shares among 3 friends' does not count the sharer: there the 3 friends are all the people.)

Watch for: Saying Yes because the story says 3 cousins.; Sharing 20 among 3 people.

T3. Mei has 18 sweets. Read the two stories.

Answer: Group; 3 bags; Share; 3 sweets

Story A tells us 6 sweets in each bag, so it is Group, and 18 in groups of 6 makes 3 bags. Story B tells us 6 bags, so it is Share, and 18 shared among 6 bags is 3 in each bag. Both answers are 3, but the meanings are different.

Watch for: Saying Share for both stories.; Answering 6 for (b) or (d).

T4. Ravi has 17 stickers. He puts 5 stickers in each bag.

Answer: 3 bags; 2 left over; 3 more

3 bags use 15 stickers, so 2 stickers are left over. One more bag needs 5 stickers, and 2 + 3 = 5, so he needs 3 more.

Watch for: Writing 2 for (c), the number left over.; Writing 4 bags.

Go further (page 4)

G1. Jenny has 12 sweets. She shares them equally among some friends. Fill in the table. Then answer (a) and (b).

Answer: 12, 6, 4, 3, 2, 1; No; 2 left over

1 friend gets all 12. 2 friends get 6 each. 3 friends get 4 each. 4 friends get 3 each. 6 friends get 2 each. 12 friends get 1 each. With 5 friends, each gets 2 and that uses 10 sweets, so 2 are left over and the answer to (a) is No.

Watch for: Writing the number of friends instead of the number each.; Saying Yes for 5 friends.

G2. Jenny shares 20 pencils equally among 4 pots. Then she puts 3 more pencils in each pot.

Answer: 5 pencils; 8 pencils; 32 pencils

20 pencils among 4 pots is 5 in each pot. Then 3 more in each pot makes 5 + 3 = 8 in each pot. 4 pots with 8 pencils is 8 + 8 + 8 + 8 = 32 pencils in all.

Watch for: Adding 3 to 20.; Stopping after 5.

G3. Wei has 20 stickers. He keeps 2 stickers. He shares the rest equally among 3 friends.

Answer: 18 stickers; 6 stickers each

20 - 2 = 18 stickers are shared. 18 shared among 3 friends is 6 each, because 3 groups of 6 is 18.

Watch for: Sharing all 20 stickers.; Writing 7 by sharing 20 among 3.

Investigation (page 5): Sharing among 4 plates

Left over: 1, 2, 3, 0, 1, 2, 3, 0. The left-over numbers 0, 1, 2, 3 repeat. Never more than 3 are left over, because with 4 left over we could give one more counter to every plate.

Look for: A child who deals out the counters for several numbers, writes the left-over numbers and notices that they repeat 1, 2, 3, 0. Credit a child who says the same numbers come back every 4 counters, or who says the left over is never as big as the number of plates. Do not mark down a child who finishes only some of the numbers.

Exit ticket (page 5)

E1. 20 sweets are shared equally among 4 children. Fill in the blank.

Answer: 5

Deal 20 sweets to 4 children one at a time: 5 each. Check: 4 groups of 5 is 20.

Watch for: Writing 4, the number of children.; Writing 24 by adding.

E2. Which story puts 12 sweets into groups of 4? Write its number in the bracket.

Answer: (2) Mei puts 12 sweets into bags of 4.

In story 2 the sweets are put into bags of 4, so they are in groups of 4 and there are 3 bags. Story 1 shares among 4 friends (3 each). Stories 3 and 4 do not make groups.

Watch for: Choosing story 1 because it has a 4 in it.; Choosing story 3 because it takes away 4.

E3. Ali puts 18 cards into piles. Each pile has 3 cards.

Answer: 6 piles; 6 × 3 = 18

Count in jumps of 3 to 18: 3, 6, 9, 12, 15, 18 is 6 jumps, so 6 piles. Check: 6 × 3 = 18.

Watch for: Writing 3, the number in each pile.; Writing 15 by stopping early.

Sneak-peek puzzles (not in the workbook)

1. 3
2. Mei. Ali puts 3 sweets in each bag (4 bags). Mei shares 12 among 3 bags, so each bag has 4.

Answer key: Lesson 8, Times Stories

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Mei has 3 bags. Each bag has 5 sweets. Write the number in each box of the strip. Then fill in the blank.

Answer: 5, 5, 5 in the boxes; 15 sweets

There are 3 bags, so 3 boxes, and each box is 5 sweets. 5 + 5 + 5 = 15, so Mei has 15 sweets in all.

Watch for: Writing 3 in the boxes.; Adding 3 + 5 = 8.

S2. Ben has 4 boxes. Each box has 3 pencils. How many pencils does Ben have in all?

Answer: 4 groups of 3 = 12 (3 groups of 4 = 12 is also right); 12 pencils

4 boxes with 3 pencils in each box: 3 + 3 + 3 + 3 = 12. So 4 groups of 3 = 12 and Ben has 12 pencils. Either order gives the right total: if a child writes 3 groups of 4 = 12, ask 'how many boxes?'.

Watch for: Writing 4 + 3 = 7 by adding.; Drawing 3 boxes with 4 in each box (the total is right, so ask 'how many boxes?').

S3. Siti has 5 bags. Each bag has 2 sweets. Which strip shows Siti's sweets? Write its number in the bracket.

Answer: Strip 2

Strip 2 has 5 boxes with 2 in each box: 5 bags of 2 sweets. Strip 1 has 2 boxes of 5. Strip 3 has 5 boxes but the last box has 1, so the boxes are not equal.

Watch for: Choosing strip 1 because it has the same numbers.; Not noticing that the last box of strip 3 is smaller.

S4. Ali has 3 boxes with 5 pens in each box. He also has 2 loose pens. How many pens does Ali have in all?

Answer: 15 pens in the boxes; 17 pens in all

Step 1: 3 boxes of 5 pens is 5 + 5 + 5 = 15. Step 2: 15 and 2 loose pens make 17 pens in all.

Watch for: Writing 5 + 3 + 2 = 10.; Forgetting the loose pens.

S5. Mei has 2 bags. Each bag has 6 sweets. Draw a strip. Then fill in the blank.

Answer: A strip of 2 boxes with 6 in each box; 12 sweets

2 bags with 6 sweets in each bag: draw 2 equal boxes and write 6 in each. 6 + 6 = 12, so Mei has 12 sweets in all.

Watch for: Drawing 6 boxes with 2 in each.; Drawing boxes that are not the same size.

S6. Ravi has 4 bags with 3 marbles in each bag. He loses 2 marbles. How many marbles does Ravi have now?

Answer: 12 marbles in the bags; 10 marbles now

Step 1: 4 bags of 3 marbles is 3 + 3 + 3 + 3 = 12. Step 2: he loses 2, so 12 - 2 = 10 marbles are left.

Watch for: Adding the 2 marbles.; Taking 2 away from 4 or from 3.

Think harder (page 3)

T1. Jenny has 3 boxes with 6 pens in each box and 5 loose pens. She gives 8 pens away.

Answer: 18 pens; 23 pens; 15 pens left

Step 1: 3 boxes of 6 is 6 + 6 + 6 = 18. Step 2: 18 and 5 loose pens make 23. Step 3: 23 take away 8 is 15 pens left.

Watch for: Stopping at 23.; Taking away 8 from 18 before adding the loose pens.

T2. Mei has 7 marbles. Ali has twice as many marbles as Mei.

Answer: 14 marbles; 21 marbles together

Mei's strip is 7. Ali has twice as many: 7 and 7, which is 14. Together they have 7 + 14 = 21 marbles.

Watch for: Writing 9 (7 + 2).; Writing 14 for together.

T3. Mei has 4 stickers. Ravi has twice as many stickers as Mei. Jenny says Ravi has 6 stickers.

Answer: No; 8 stickers

Mei has 4, so Ravi has two strips of 4: 4 + 4 = 8 stickers. Jenny added 2 to 4 to get 6, but twice as many is not 2 more. So Jenny is not correct.

Watch for: Saying Yes because 4 and 2 make 6.; Writing 4 or 12.

T4. Ali has twice as many cards as Mei. Ali has 24 cards. Ali gives some cards to Mei. Then they have the same number of cards.

Answer: Mei 12 cards; Ali gives 6 cards

Ali has two strips of 12, so Mei has 12. The gap is 12. Ali gives half of the gap: two equal parts of 12 are 6 and 6, so he gives 6. Then Ali has 24 - 6 = 18 and Mei has 12 + 6 = 18. They have the same number.

Watch for: Giving all 12 cards.; Writing 12 for (b).

Go further (page 4)

G1. Wei puts 3 stickers on each of 4 cards. He puts 2 stickers on each of 5 other cards. He also has 4 stickers in his hand.

Answer: 12 stickers; 10 stickers; 26 stickers at first

4 cards with 3 stickers: 3 + 3 + 3 + 3 = 12. 5 cards with 2 stickers: 2 + 2 + 2 + 2 + 2 = 10. 12 + 10 + 4 = 26 stickers at first (the 4 in his hand count too).

Watch for: Forgetting the 4 stickers in his hand.; Using 3 for both sets of cards.

G2. Siti has 5 marbles. Ravi has 3 times as many marbles as Siti.

Answer: 15 marbles; 20 marbles altogether

Siti's strip is 5. Ravi has 3 times as many: 5 + 5 + 5 = 15. Together: 5 + 15 = 20 marbles.

Watch for: Writing 8 (5 + 3).; Writing 15 for altogether.

G3. Kumar has 3 times as many pens as Lily. They have 24 pens together.

Answer: Lily 6 pens; Kumar 18 pens

Lily has 1 box and Kumar has 3 boxes, so there are 4 equal boxes. 24 pens in 4 equal boxes is 6 in each box (6 + 6 + 6 + 6 = 24). Lily has 6 and Kumar has 18. Check: 3 × 6 = 18 and 6 + 18 = 24.

Watch for: Writing 8 (24 shared into 3).; Writing 3 times 24.

Investigation (page 5): Twice, and twice again

Mei 3, Ali 6, Ben 12, Jenny 24. Each child has twice as many as the child before, so each strip is twice as long as the one before. The numbers double each time: 3, 6, 12, 24.

Look for: A child who draws or builds each strip as two copies of the strip before, writes 3, 6, 12, 24 and notices that each number is double the last. Credit a child who says the strips get twice as long, or who checks 12 + 12. Do not mark down a child who finds only Ali and Ben.

Exit ticket (page 5)

E1. Ali has 4 bags. Each bag has 5 sweets. Which number sentence finds how many sweets Ali has in all? Write its number in the bracket.

Answer: (2) 4 × 5

4 bags with 5 sweets in each bag is 4 groups of 5, which is 4 × 5. Adding 4 + 5 joins the numbers of bags and sweets. 5 - 4 finds a gap.

Watch for: Choosing 4 + 5.; Choosing 5 - 4.

E2. Lily has 2 bags with 5 marbles in each bag. She also has 3 loose marbles. How many marbles does Lily have in all?

Answer: 13

Step 1: 2 bags of 5 marbles is 10. Step 2: 10 and 3 loose marbles make 13 marbles in all.

Watch for: Writing 10 (forgetting the loose marbles).; Writing 2 + 5 + 3 = 10.

E3. Ravi has 3 bags with 4 sweets in each bag. He eats 2 sweets. How many sweets does Ravi have now?

Answer: 10

Step 1: 3 bags of 4 sweets is 12. Step 2: he eats 2, so 12 - 2 = 10 sweets are left.

Watch for: Adding the 2 sweets: 14.; Writing 12 (forgetting the 2 he eats).

Sneak-peek puzzles (not in the workbook)

1. 12
2. 15. Ben has twice as many as Mei: 5 and 5, which is 10. Mei has 5 and Ben has 10, so together they have 15.

Answer key: Lesson 9, Picture Equations

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. A cup hides some counters. 3 counters are outside the cup. There are 11 counters in all. Fill in the blank.

Answer: 8

11 - 3 = 8, so the cup holds 8 counters. Check: 8 + 3 = 11.

Watch for: Writing 14 (11 + 3).; Writing 11 (the total).

S2. Each shape is a number. Find the number for the circle.

Answer: 6

The total is 10 and the square is 4. 10 - 4 = 6, so the circle is 6. Check: 6 + 4 = 10.

Watch for: Writing 14 (adding 10 and 4).; Writing 10 (the total).

S3. Each shape is a number. Find the number for the heart.

Answer: 8

8 + 8 = 16, so each heart is 8.

Watch for: Writing 16 (the total).; Writing 32 (adding the total twice).

S4. Each shape is a number. Find the number for the star.

Answer: 5

5 + 5 + 5 = 15, so each star is 5.

Watch for: Writing 3 (counting the stars).; Writing 12 (15 - 3).

S5. Each shape is a number. The square is 4. Fill in the blanks.

Answer: 10; 5

14 - 4 = 10, so the 2 triangles make 10. The triangles are the same, so each is 5. Check: 5 + 5 + 4 = 14.

Watch for: Writing 14 for each triangle.; Writing 10 for each triangle.

S6. Ben says the circle is 7. Look at the picture.

Answer: No; 6

7 + 7 = 14, but the line says 12. So Ben is not correct. 6 + 6 = 12, so the circle is 6.

Watch for: Saying Yes because 7 is close to 6.; Writing 12 for the circle.

S7. Each bag has the same number of sweets. 3 sweets are outside the bags. There are 13 sweets in all. Fill in the blanks.

Answer: 10; 5

13 - 3 = 10 sweets are in the 2 bags. 5 + 5 = 10, so each bag has 5 sweets. Check: 5 + 5 + 3 = 13.

Watch for: Writing 13 for each bag.; Writing 6 (13 shared between 2 bags, ignoring the loose sweets).

Think harder (page 3)

T1. Each shape is a number. Find the number for each shape. Show your working.

Answer: square 6; heart 7

Line 1 has one more square than line 2, and 19 is 6 more than 13, so the square is 6. Then 6 + heart = 13, so the heart is 7. Check: 6 + 6 + 7 = 19.

Watch for: Writing 19 and 13 as the answers.; Finding the square but forgetting the heart.

T2. Kumar says the heart is 8 and the star is 7. Look at the picture. Decide if each statement is True or False.

Answer: True; False

Line 1: 8 + 7 = 15, which matches. Line 2: 8 + 8 + 7 = 23, but the line says 21. Kumar's numbers work in line 1 only.

Watch for: Checking only line 1.; Saying line 2 is true because 23 is close to 21.

T3. Each shape is a number. Find the number for each shape.

Answer: heart 3; circle 5

The 2 hearts make 6, so the heart is 3. Then 3 × circle = 15 means 3 groups of 5 make 15, so the circle is 5. Check: 3 × 5 = 15.

Watch for: Writing 6 for the heart.; Writing 15 for the circle.

Go further (page 4)

G1. Each shape is a number. Find the number for each shape.

Answer: circle 5; heart 7; star 11

Circle × circle = 25, so the circle is 5 (5 groups of 5). Then heart + 5 = 12, so the heart is 7. Then star - 7 = 4, so the star is 11. Check: 11 - 7 = 4.

Watch for: Writing 12 for the heart.; Taking 4 away from 7 and writing 3 for the star.

G2. Each shape is a number. Fill in the blanks.

Answer: 14; star 8; heart 6

Both lines together have 3 stars and 3 hearts, and 22 + 20 = 42. 3 stars and 3 hearts make 42, so one star and one heart make 14. Then 22 - 14 = 8 is a star, and 14 - 8 = 6 is a heart.

Watch for: Writing 22 and 20 as the answers.; Writing 7 for both shapes.

G3. Each shape is a number. The heart is greater than the triangle. Find the number for each shape.

Answer: heart 5; triangle 4

The pairs that add to 9 are 1 and 8, 2 and 7, 3 and 6, 4 and 5. Only 4 and 5 multiply to 20. The heart is greater, so the heart is 5 and the triangle is 4.

Watch for: Writing 4 for the heart.; Using 3 and 6 (they add to 9 but multiply to 18).

Investigation (page 5): One line or two?

Only circle 4 and triangle 6 fit both lines. One line alone has many pairs that fit (1 and 9, 2 and 8, 3 and 7, 4 and 6, 5 and 5 and more; accept 5 and 5 for line 1). 5 and 5 fails line 2 (5 + 5 + 5 = 15, not 14). The second line is the same as the first with one more circle, and 14 is 4 more than 10, so the circle is 4.

Look for: A child who tries pairs for both lines, finds that only one pair works, or says one line is not enough to find two numbers. Credit any clear reason. Do not mark a child down for stopping at one pair.

Exit ticket (page 5)

E1. Each shape is a number. The triangle + the triangle = 18. Find the number for the triangle.

Answer: 9

9 + 9 = 18, so each triangle is 9.

Watch for: Writing 18 (the total).; Writing 2 (counting the triangles).

E2. The heart + the square = 13. The square is 5. Find the number for the heart. Choose the correct answer. Write its number in the bracket.

Answer: (4) 8

13 - 5 = 8, so the heart is 8. Check: 8 + 5 = 13.

Watch for: Choosing (1) 5, the number of the square.; Choosing (2) 6 or (3) 7 after a counting slip.

E3. Each shape is a number. The star is 5. Find the number for each circle.

Answer: 7

19 - 5 = 14, so the 2 circles make 14. 7 + 7 = 14, so each circle is 7. Check: 7 + 7 + 5 = 19.

Watch for: Writing 14 for each circle.; Writing 19 for each circle.

Sneak-peek puzzles (not in the workbook)

1. 5

2. 3. The two stars make 14, so one star is 7. The heart and 7 make 10, so the heart is 3.

Answer key: Lesson 10, Find One First

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. 4 pens cost $20. How much does 1 pen cost?

Answer: $5

$20 shared equally among 4 pens is $5 each. Check: 4 groups of $5 make $20.

Watch for: Writing 24 (adding 20 and 4).; Writing 80 (multiplying).; Writing 20 (the total).

S2. 1 pen costs $3. How much do 4 pens cost? Fill in the blanks.

Answer: 12; $12

4 groups of $3 is 3 + 3 + 3 + 3 = 12, so 4 pens cost $12.

Watch for: Writing 7 (adding 3 and 4).; Writing 3 (the cost of 1 pen).

S3. 6 buns cost $12. How much does 1 bun cost? Look at the strip.

Answer: $2

$12 shared equally among 6 buns is $2 each. Check: 6 groups of $2 make $12.

Watch for: Writing 6 (the number of buns).; Writing 72 (multiplying).

S4. 3 books cost $15. Fill in the blanks. Look at the strip.

Answer: $5; $10

$15 shared equally among 3 books is $5 each. 2 books cost 2 groups of $5, which is $10. Check: 3 groups of $5 make $15.

Watch for: Writing 15 for 2 books.; Writing 7 (adding 5 and 2).; Stopping after the first part.

S5. 4 pens cost $12. Lily says 1 pen costs $4. Look at the picture.

Answer: No; $3

4 groups of $4 make $16, not $12. So Lily is not correct. $12 shared equally among 4 pens is $3 each. Check: 4 groups of $3 make $12.

Watch for: Saying Yes because 4 is the number of pens.; Writing 12 for 1 pen.

S6. Each book costs $3. Kumar buys 5 books. How much do the 5 books cost?

Answer: $15

5 groups of $3: 3 + 3 + 3 + 3 + 3 = 15, so the 5 books cost $15.

Watch for: Writing 8 (adding 3 and 5).; Writing 3.

Think harder (page 3)

T1. 3 books cost $12. Fill in the blanks. Look at the strip.

Answer: $4; $28

$12 shared equally among 3 books is $4 each. 7 books cost 7 groups of $4, which is $28. Check: 3 groups of $4 make $12.

Watch for: Writing 84 (12 × 7).; Writing 19 (12 + 7).; Stopping after the first part.

T2. A dozen buns cost $24. Fill in the blanks.

Answer: $2; $6

A dozen is 12, so 12 buns cost $24. $24 shared equally among 12 buns is $2 each. 3 buns cost 3 groups of $2, which is $6.

Watch for: Using 24 shared among 2 buns.; Writing 24 for 3 buns.

T3. Wei pays $18 for 6 such cups. Fill in the blanks.

Answer: $3; $12

$18 shared equally among 6 cups is $3 each. 4 such cups cost 4 groups of $3, which is $12.

Watch for: Writing 18 for 4 cups.; Writing 7 (3 + 4).

T4. 5 pens cost $15. Ben has $12. Fill in the blanks.

Answer: $3; 4 pens

$15 shared equally among 5 pens is $3 each. $12 holds 4 groups of $3, so Ben can buy 4 pens.

Watch for: Writing 5 pens (the number in the story).; Writing 36 (12 × 3).

Go further (page 4)

G1. 2 hats cost $10. 3 scarves cost $18. Jenny buys 1 hat and 2 scarves. Fill in the blanks.

Answer: $5; $6; $17

$10 shared equally among 2 hats is $5 each. $18 shared equally among 3 scarves is $6 each. 2 scarves cost $12. 1 hat and 2 scarves cost $5 + $12 = $17.

Watch for: Adding 10 and 18.; Counting 1 scarf instead of 2.

G2. 6 pens cost $18. Mei has $20. She buys 4 pens. Fill in the blanks.

Answer: $3; $12; $8

$18 shared equally among 6 pens is $3 each. 4 pens cost $12. $20 - $12 = $8 is left.

Watch for: Taking 18 away from 20.; Forgetting to take the cost away from $20.

G3. Shop A sells 4 pens for $12. Shop B sells 5 pens for $20. Which shop sells 1 pen for less?

Answer: $3; $4; Shop A

Shop A: $12 shared among 4 pens is $3. Shop B: $20 shared among 5 pens is $4. $3 is less than $4, so Shop A sells 1 pen for less.

Watch for: Choosing Shop B because 20 is more.; Comparing $12 and $20 without finding 1 pen.

Investigation (page 5): Which pack costs less?

No. In both packs 1 pen costs $3 ($6 shared between 2 pens, and $15 shared among 5 pens), so 1 pen does not cost less in the bigger pack. Finding the price of one lets us compare packs of different sizes.

Look for: A child who finds $3 for both packs, or who says 1 pen costs the same in both. Credit any clear reason. A child who says the bigger pack costs more is right about the total but has not compared one pen.

Exit ticket (page 5)

E1. 3 pencils cost $9. How much does 1 pencil cost?

Answer: $3

$9 shared equally among 3 pencils is $3 each.

Watch for: Writing 12 (adding 9 and 3).; Writing 27 (multiplying).

E2. Each sticker costs $2. How much do 5 stickers cost? Write its number in the bracket.

Answer: (2) $10

Each sticker costs $2, so 5 stickers cost 5 groups of $2, which is $10.

Watch for: Choosing (1) $7 by adding 2 and 5.; Choosing (3) $3 by taking 2 from 5.

E3. 4 buns cost $12. Fill in the blanks.

Answer: $3; $9

$12 shared equally among 4 buns is $3 each. 3 buns cost 3 groups of $3, which is $9.

Watch for: Writing 12 for 3 buns.; Writing 7 (adding 3 and 4).

Sneak-peek puzzles (not in the workbook)

1. $2
2. $10. 3 apples cost $6, so 1 apple costs $2. 5 apples cost 5 groups of $2, which is $10.